

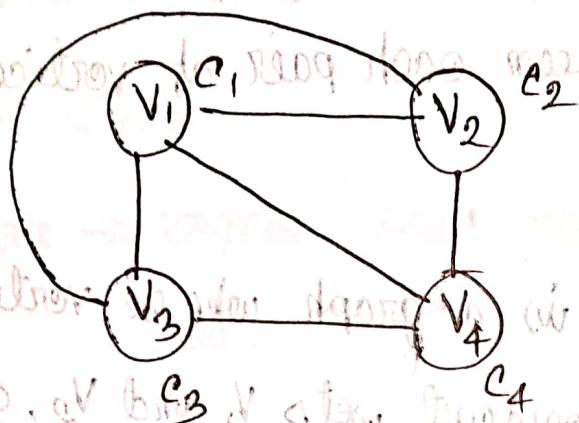
5 no - (a)

- (i) Simple Graph - An undirected graph in which there is at most one edge between each pair of vertices, and there are no loops.
- (ii) Bipartite Graph - It is a graph whose vertices V can be divided into two independent sets V_1 and V_2 , and every edge of the graph connects one vertex in V_1 to one vertex in V_2 .
- (iii) Multigraph - It is a graph which is permitted to have multiple edges, that is edges that have the same end nodes.
- (iv) Weighted Graph - It is a graph in which weights or numerical values are assigned to each of the edges.

5 no - (b)

Four color theorem: The chromatic number of a planar graph is no greater than four. Chromatic number denotes minimum number of colors needed to colour a graph such that no two adjacent vertices has the same color.

Example: K_4 is a planar graph with four vertices



Here the chromatic number is 4: c_1, c_2, c_3 and c_4 denotes four different colors.

If G is a bipartite graph with vertex sets V_1 and V_2 every step along a walk takes you either from V_1

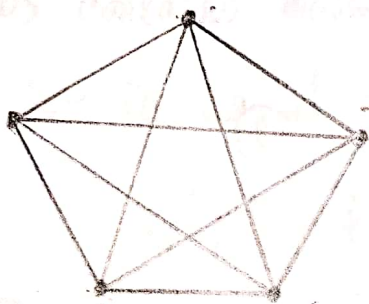
to V_2 or from V_2 to V_1 . To end up where you started, there has to be an even number of steps.

Suppose every cycle of G is even. Let v_0 be any vertex. For each vertex v in the same component

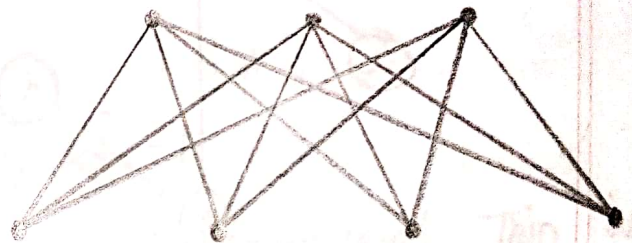
as v_0 let $d(v)$ be the length of the shortest path from v_0 to v . Color red every vertex in C_0 whose distance from v_0 is even, and color the other vertices of C_0 blue. Do the same for each

component of G . If G had any edge between two red vertices or between two blue vertices, it would have an odd cycle. Thus, G is bipartite, the red vertices and blue vertices being the two parts.

5(c)



K_5



$K_{3,4}$

5(d)

Isomorphism: Two graphs which contain the same number of graph vertices connected in the same way are said to be isomorphic. The mechanism is called Isomorphism.

Adjacency List:

A: B, D

A': B, C, D

B: A, D

B': A, D

C: D

C': A, D

D: A, B, C

D': A, B, C.

Between two graphs, the vertices are the same but the edges of these graphs aren't the same. That's why they are not isomorphic.

7(a)

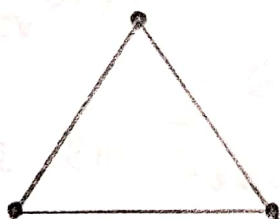
The generalized Pigeonhole Principle: If N objects are placed into k boxes, then there is at least one box containing at least $\lceil N/k \rceil$ objects.

Proof: We use a proof by contraposition. Suppose that none of the boxes contains more than $\lceil N/k \rceil - 1$ objects. Then the total number of objects is at most

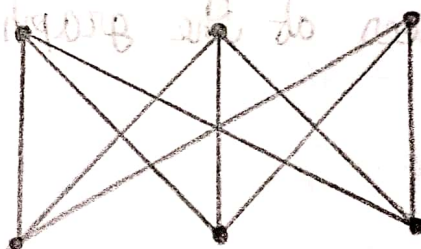
$$k(\lceil N/k \rceil - 1) < k(\lceil N/k \rceil) = N,$$

where the inequality $\lceil N/k \rceil < \lceil N/k \rceil + 1$ has been used. This is a contradiction because there are a total of n objects.

7(c)



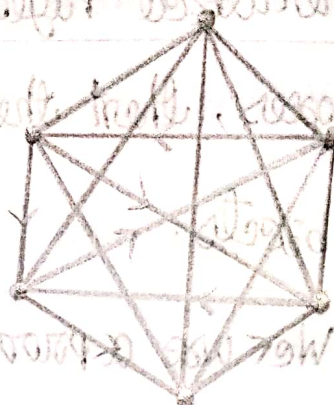
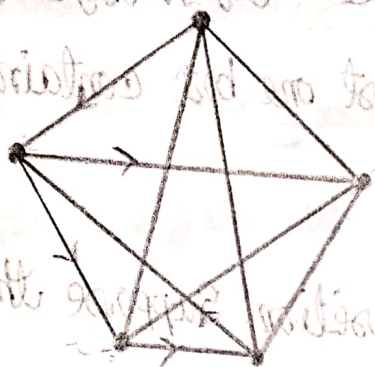
3-regular graph



$K_{3,3}$

In 3-regular graph, no edge crosses each other. But Any attempt to draw $K_{3,3}$ in the plane with no edges crossing is denoted. So, we can say that 3-regular graph is a planar graph.

(7(d))



The vertices of K_5 all have even degree, so an Eulerian circuit exists. K_6 doesn't have all even degree.

A directed graph is called strongly connected if there is a path in each direction between each pair of vertices of the graph.

5(b)

Complete Graph - It is a graph, simple undirected graph, in which every pair of distinct vertices is connected by a unique edge.

Connected Graph - If every pair of vertices in the graph is connected, it is called connected graph.

5(c)

Adjacency Matrix: An adjacency matrix is a square matrix used to represent a finite graph. The elements of the matrix indicate whether pairs of vertices are adjacent or not in the graph.

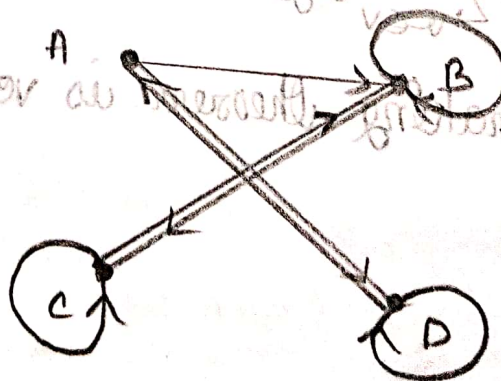
Adjacency List

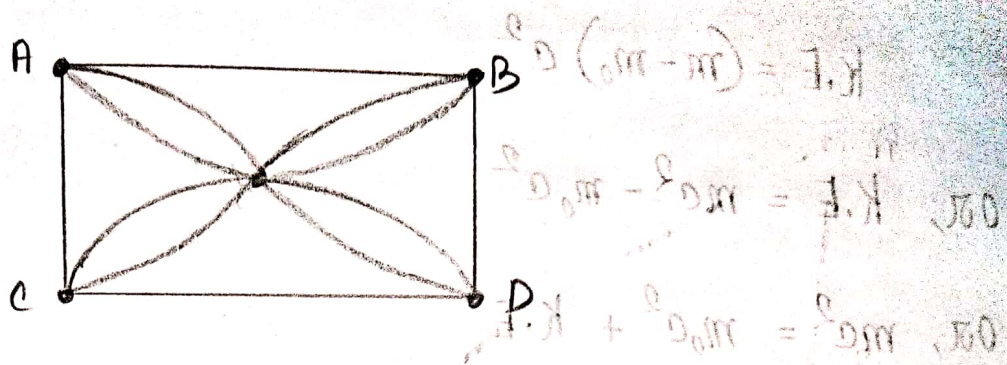
A : B, D

B : B, C

C : B, C

D : A, D





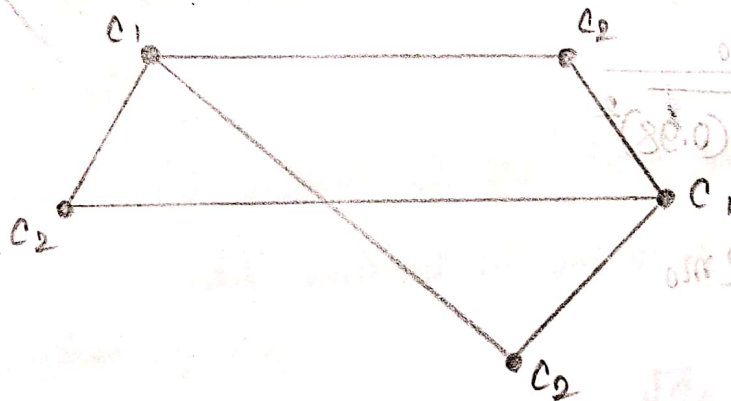
Planar Graph:

7(c)

It is a graph where no edge crosses each other.

In the graph fig-7(c) edges crosses each other, so it is not a planar graph.

7(d)



The chromatic number of the graph is 2: c_1 and c_2 .